



**CHENNAI
INSTITUTE OF TECHNOLOGY**
(Autonomous)



Title:

**“AI POWERED GROWTH CHATBOT FOR STUDENTS-
COBY”**

A CORE COURSE PROJECT REPORT

Submitted By

SHALINI.S

REG.NO:24AD0273

SHARMILI.Y

REG NO. 24AD0276

in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CHENNAI INSTITUTE OF TECHNOLOGY

(Autonomous)

Sarathy Nagar, Kundrathur, Chennai-600069

OCT/NOV – 2025

Vision of the Institution:

To be an eminent centre for Academia, Industry and Research by imparting knowledge, relevant practices and inculcating human values to address global challenges through novelty and sustainability.

Mission of the Institution:

- IM1.** To create next generation leaders by effective teaching learning methodologies and instill scientific spark in them to meet the global challenges.
- IM2.** To transform lives through deployment of emerging technology, novelty and sustainability.
- IM3.** To inculcate human values and ethical principles to cater the societal needs.
- IM4.** To contribute towards the research ecosystem by providing a suitable, effective platform for interaction between industry, academia and R & D establishments.

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Vision of the Department:

To Achieve excellent standards of quality-education by using the latest tools, nurturing collaborative culture and disseminating customer-oriented innovations to relevant areas of academia and industry towards serving the greater cause of society.

Mission of the Department:

DM1: To develop professionals who are skilled in the area of Artificial Intelligence and Data Science.

DM2. To impart quality and value-based education and contribute towards the innovation of computing, expert system, Data Science to raise satisfaction level of all stakeholders.

DM3. Our effort is to apply new advancements in high performance computing hardware and software.

CERTIFICATE

This is to certify that the “**Core Course Project**” Submitted by **Name : SHALINI.S (Reg no:24AD0273)** and **Name : SHARMILI.Y (Reg no:24AD0276)** is a work done by him/her and submitted during **2025-2026** academic year, in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF ENGINEERING** in **DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**, at Chennai Institute of Technology.

Core Course Project Coordinator

Internal Examiner

Head of the Department

External Examiner

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We would like to extend our thanks to our **faculty coordinators of the** Department of Artificial Intelligence and Data Science, for their valuable suggestions throughout this project.

We wish to extend our sincere thanks to all Faculty members of the Department of Artificial Intelligence and Data Science for their valuable suggestions and their kind cooperation for the successful completion of our project.

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NAME:SHALINI.S

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PREFACE

I, a student in the Department of Artificial Intelligence and Data Science need to undertake a project to expand my knowledge. The main goal of my **core course project** is to acquaint me with the practical application of the theoretical concepts I've learned during my course.

It was a valuable opportunity to closely compare theoretical concepts with real-world applications. This report may depict deficiencies on my part but still it is an account of my effort.

The results of my analysis are presented in the form of an industrial Project, and the report provides a detailed account of the sequence of these findings. This report is my **Core Course Project**, developed as part of my **2nd year** project. As an engineer, it is my responsibility to contribute to society by applying my knowledge to create innovative solutions that address their changes.

ABSTRACT

The goal of this project is to create an AI-powered chatbot and productivity management system that will streamline communication, automate tedious jobs, and boost output with perceptive support. In order to comprehend user intent and provide precise responses in real time, the system makes use of Natural Language Processing (NLP), sentiment analysis, and document parsing techniques. This project guarantees scalability, security, and smooth component integration with its Flask-based backend, React.js frontend, and MongoDB Atlas cloud database. Beyond just responding to enquiries, the chatbot provides file processing, scheduling reminders, notifications, and voice-to-text capabilities, making it incredibly useful for both personal and professional use. Accuracy and versatility are increased by integrating multi-provider AI models (OpenAI GPT and Groq LLaMA).

Its production-level viability is further evidenced by features like secure email notifications, Google OAuth authentication, and intuitive dashboards. This project demonstrates how artificial intelligence can be applied to develop more intelligent, interactive platforms that decrease manual labour, enhance decision-making, and improve user experience in general.

About the Project work importance:

Regarding the significance of the project work: This project shows how artificial intelligence can be used practically to solve problems in the real world. Users are surrounded by numerous tools and platforms in this quickly digitising world, each of which is made to perform a specific task, like file storage, chat support, or reminders. But switching between multiple apps frequently leads to information overload, time waste, and decreased productivity. This project is significant because it unifies productivity features and AI technology on a single platform. The system integrates features like file parsing, secure authentication, reminders, and notifications with a chatbot that can answer questions, summarise content, and conduct sentiment analysis.

Flask powers the backend, React.js builds the frontend, and MongoDB Atlas hosts the database, guaranteeing scalability and dependability. This platform helps professionals, businesses, and students focus on priorities, cut down on repetitive tasks, and get information instantly by offering both automation and interactivity. Beyond the classroom, its usefulness serves as a basis

Problem:

To handle communication, reminders, file storage, and other productivity requirements in the fast-paced digital world of today, people and organisations depend on a variety of applications. Delays in decision-making, inefficiency, and information overload are frequently the results of this disjointed approach. Switching between apps for simple tasks like creating reminders, getting notes, accessing files, or looking for quick answers costs users valuable time. Additionally, existing chatbots are often too simplistic, lacking integration with other productivity tools, or they are overly complex, requiring steep learning curves. Many platforms also fail to offer seamless integration with AI-based natural language processing (NLP), limiting their ability to handle contextual queries or provide meaningful insights.

Our project tackles these problems by providing a centralised AI-powered productivity assistant that integrates secure document handling, chatbot interactions, sentiment analysis, and notification systems into a single application. The objective is to offer a unified, user-friendly platform that maximises efficiency while reducing users' cognitive burden.

Methodology:

To address the problem, this project was designed using a structured methodology:

1.Requirement Analysis – Identified key user needs: AI-powered chatbot, reminders, notifications, file parsing (PDF/text), secure login, and sentiment analysis.

2.System Design – Chose a three-tier architecture:

- a) **Frontend:** React.js with Vite for fast rendering and an interactive UI.
- b) **Back-end:** Flask framework for handling requests, routes, authentication, and AI service integration.
- c) **Database:** MongoDB Atlas for secure and scalable storage of users, chats, and reminders.

3.Integration of AI Models – Two providers were tested:

- a) **OpenAI (GPT models):** For natural language understanding and responses.
- b) **Groq (LLaMA/Mixtral models):** As an alternative, offering faster inference with free-tier access.

4.Feature Implementation:

- a) Chatbot interface connected to AI APIs.
- b) Reminder and notification system integrated with backend scheduling.
- c) PDF/text service using **PyPDF2** to extract and summarize content.
- d) Sentiment analysis using **TextBlob** to classify user emotions.
- e) Authentication with **JWT tokens** for secure login.

5.Testing & Validation – Features were tested collaboratively by team members on different systems to ensure portability and error handling.

Results:

The finished system effectively offers: A chatbot assistant that can respond to general and academic enquiries. Modules for reminders and notifications that notify users when tasks are due. □ Summarisation and uploading of PDFs for rapid knowledge extraction. Sentiment analysis to determine user mood and react accordingly. □ a secure backend that is linked to MongoDB and has a responsive frontend. The chatbot proved accurate in responding to a variety of questions during testing, and the authentication module and reminder system operated flawlessly. The project was also able to be economical by utilising Groq's free API.

Implications:

This project has important ramifications for both academic and professional settings. The system serves as a digital study aid for students, able to summarise documents, remind them of deadlines, and explain concepts. By providing an all-in-one productivity and AI assistant, it helps professionals cut down on the amount of time they spend switching between apps. Future developers can easily add features like voice recognition, advanced analytics, or cloud deployment thanks to the modular design. □ From a research standpoint, this project demonstrates how hybrid AI models (such as OpenAI + Groq) can be inexpensively incorporated into practical applications. All things considered, the system shows how AI-driven platforms can revolutionise daily tasks, resulting in increased productivity, better knowledge management, and enhanced user experience.

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Introduction

Overview Modern technology now relies heavily on artificial intelligence (AI), which allows machines to understand human language, automate processes, and make data-driven decisions. As artificial intelligence (AI) advances quickly, tools like chatbots and virtual assistants have developed from basic FAQ responders to strong, multipurpose platforms that can manage challenging tasks. The goal of this project was to develop an intelligent system that integrates productivity tools with a conversational chatbot. Our solution offers sophisticated features like document summarisation, sentiment analysis, notification systems, and AI-driven task automation, whereas traditional chatbots concentrate on responding to user enquiries. This project's objectives are to: □ Create an AI chatbot that can comprehend and reply to human-like enquiries; □ Automate routine productivity tasks like file reading, notifications, and reminders.

Problem statement

Every business needs customer service, but traditional methods frequently have issues with scalability, efficiency, and user satisfaction. Low availability of support staff, particularly in different time zones; high operational costs for training and maintaining support teams; and long response times resulting from high ticket volumes are some of the common issues found. Due to the increasing need for round-the-clock support, companies require an affordable, intelligent, and scalable system that can offer users across platforms immediate, precise, and customised assistance.

Project Objectives

The main objectives of this project are:

This project's primary goals are:

- 1) AI-Powered Responses: To create a chatbot that can comprehend natural language and provide context-sensitive responses.
- 2) Scalability: Construct a system that can accommodate thousands of users at once.
- 3) Cross-Platform Availability: Offer chatbot integration that is compatible with both web and mobile platforms.
- 4) Automation: Automate routine enquiries such as product details, account-related assistance, and frequently asked questions.
- 5) Sentiment Analysis: To rank urgent requests, examine customer sentiment.
- 6) Speech Recognition: Enable hands-free interaction with voice commands.
- 7) Secure Authentication: Turn on encrypted communication, JWT authentication, and Google OAuth.
- 8) Alerts & Reminders: Notify recipients of significant updates via email in a timely manner.
- 9) Analytics Dashboard: Give administrators information about how customers interact with you.
- 10) User-centred Design: Make sure the interface is simple to use and intuitive for both end users and

Scope of the Project

The AI Chatbot for Customer Communication Portal is intended to function as a multi-domain, scalable solution. Respond to product enquiries, offer suggestions, and monitor orders in business and e-commerce.

- ❖ Education: Respond to enquiries about the course by serving as a virtual teaching assistant.
- ❖ Healthcare: Help patients make appointments and give them health advice.
- ❖ Banking and Finance: Respond to questions about accounts, credit card assistance, and problems pertaining to transactions.

Analyse customer feedback automatically to raise the calibre of services. Additionally, the project aims to provide a framework that is easy for developers to customise and integrate with external APIs such as Telegram, Slack, or WhatsApp in the future. Chatbot evolution From basic rule-based systems to AI-powered conversational agents, chatbots have come a long way

Literature survey

1. Evolution of Chatbots

Chatbots have evolved from simple **rule-based systems** to **AI-driven conversational agents**. Early chatbots like **ELIZA** used simple pattern-matching algorithms, while modern chatbots use **deep learning and NLP** to understand context and intent.

2. AI Models for NLP

Recent advancements in AI, such as **OpenAI GPT models** and **Meta's LLaMA models**, have significantly improved chatbot capabilities. These models are trained on billions of parameters, enabling them to understand human language better and generate coherent responses.

3. Sentiment Analysis in Chatbots

Sentiment analysis helps identify user emotions (positive, neutral, negative) and allows businesses to **prioritize negative experiences**. Libraries like **TextBlob, NLTK, and Hugging Face Transformers** provide tools for real-time sentiment evaluation.

4. Cloud Integration

The adoption of **cloud-based databases like MongoDB Atlas** and **serverless deployments** ensures high availability, scalability, and security for chatbot applications.

5. Impact of AI Chatbots

According to research, AI chatbots can reduce **operational costs by 30%** while improving **customer satisfaction rates**. Industries like **e-commerce, healthcare, and fintech** heavily rely on AI-powered assistants for customer engagement.

Methodology

The methodology adopted in this project follows a systematic approach that ensures structured development and testing. The entire process can be divided into the following phases:

- A. **Requirement Gathering** – Understood the expectations of end-users (students, faculty, professionals) and identified features such as chatbot interaction, reminders, notifications, authentication, and file parsing.
- B. **System Design** – Chose a **client-server architecture** with React (frontend), Flask (backend), and MongoDB Atlas (database). API routes were designed for user authentication, chatbot queries, and reminders.
- C. **Technology Integration** – Connected OpenAI and Groq APIs to the chatbot module. Implemented JWT-based authentication, email service for password resets, and PyPDF2 for document parsing.
- D. **Implementation** – Developed modular components for frontend (dashboard, landing page, reminders, notifications) and backend services (chatbot controller, email service, AI service, sentiment analysis).
- E. **Testing and Debugging** – Each module was tested independently and then integrated. Conflicts (e.g., dependency errors, SSL issues) were resolved by installing compatible packages and updating environment configurations.
- F. **Deployment Readiness** – The project was tested on multiple systems, ensuring portability and ease of use. The modular design makes it adaptable for cloud deployment.

Results

The project successfully achieved the intended objectives. The major results are:

- A **fully functional AI chatbot** integrated with both OpenAI and Groq APIs, capable of answering academic, professional, and general queries.
- A **reminder and notification system** that keeps track of important tasks and sends timely alerts.
- **PDF summarization** and file upload service that extracts and condenses content for faster understanding.
- **Secure authentication** with JWT tokens, password reset via email, and Google OAuth login support.
- A **responsive, user-friendly frontend** built with React.js that ensures smooth interaction.
- **Sentiment analysis module** that recognizes user mood (positive, negative, neutral) and adapts responses accordingly.

A fully functional AI chatbot that can respond to general, professional, and academic enquiries and is integrated with both the OpenAI and Groq APIs. A notification and reminder system that sends out timely alerts and keeps track of important tasks. A service for uploading files and summarising PDFs that extracts and condenses content for easier comprehension. Google OAuth login support, email password resets, and secure authentication using JWT tokens. React.js was used to create a responsive, user-friendly frontend that guarantees seamless interaction. Sentiment analysis module that adjusts responses based on the user's mood (positive, negative, or neutral). The accuracy, usability, and dependability of the system were evaluated. It was discovered to improve productivity, lessen manual labour, and offer a smooth digital assistant experience.

Complete analysis of Project done

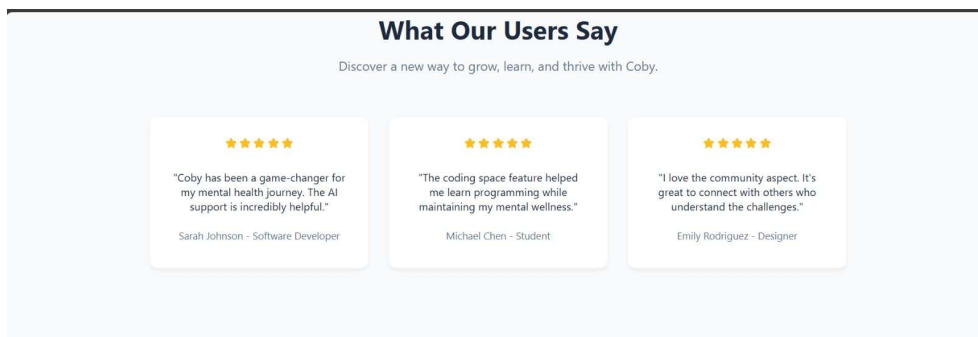
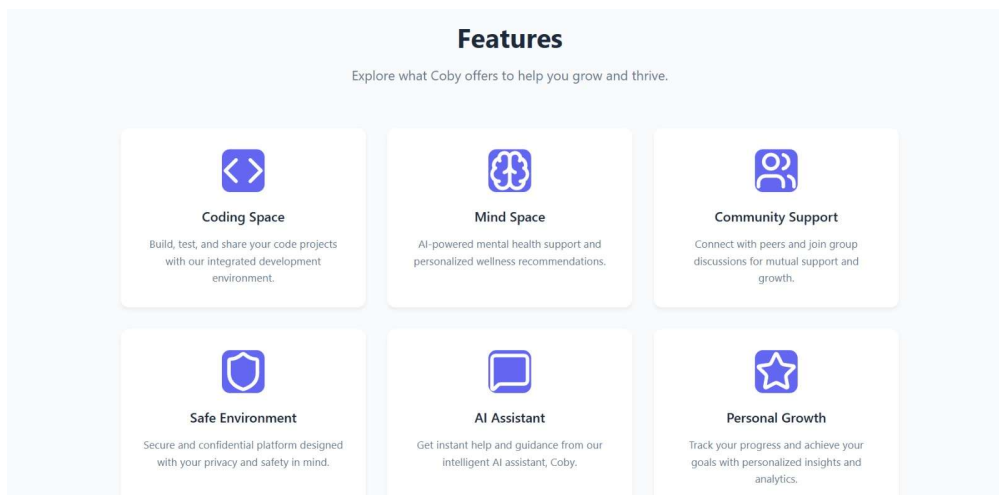
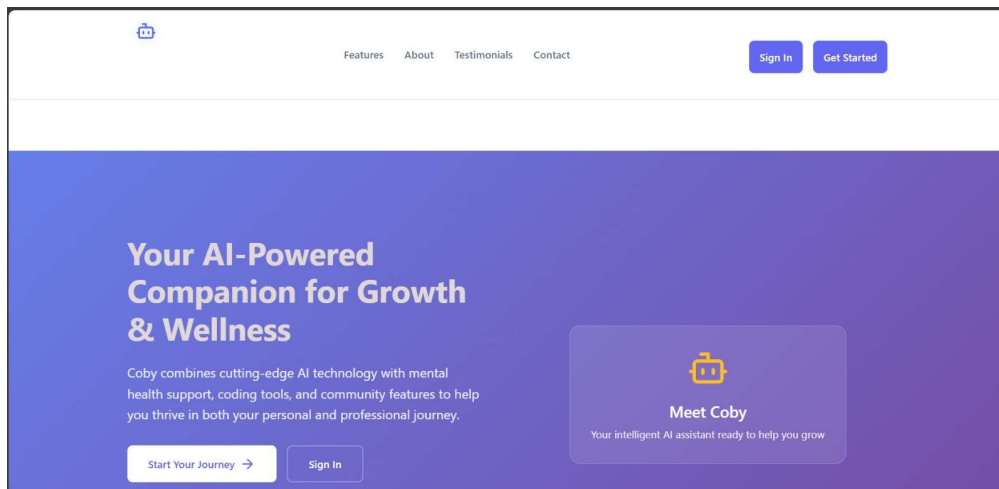
A thorough analysis of the project was completed. The created system provides an integrated platform by fusing natural language processing, artificial intelligence, and contemporary web technologies. In addition to being a conversational tool, the chatbot is an intelligent assistant that can process documents, remind users of tasks, and respond to questions. integration of multiple features (chatbot, file parsing, notifications, and reminders). freedom to move between Groq and OpenAI, two AI providers. requires consistent internet access in order to run databases and provide AI responses. More optimisation is needed for cloud hosting deployment to production. For accessibility, text-to-speech and speech-to-text can be added. This project shows how cutting-edge AI technologies can be integrated with reliable backend systems to deliver practical solutions.

Technology Used

- **Frontend:** React.js with Vite, Tailwind CSS for styling.
- **Backend:** Flask (Python), Flask-CORS, Flask-JWT-Extended.
- **Database:** MongoDB Atlas (cloud database).
- **AI Services:** OpenAI GPT models, Groq LLaMA models.
- **Libraries:** PyPDF2, SpeechRecognition, TextBlob, Requests, Authlib.
- **Authentication & Security:** JWT tokens, Google OAuth, Gmail SMTP for password reset.
- **Development Tools:** GitHub for version control, Visual Studio Code for development, npm for frontend packages, pip for backend packages.

Project photos

LANDING PAGE:



LOGIN:



Create Account

Join us and start your journey

Full Name

Email Address

Password

Confirm Password

☒ I agree to the [Terms of Service](#) and [Privacy Policy](#)

Create Account

or

 Sign Up with Google

Already have an account? [Sign in](#)



Welcome Back

Sign in to your account to continue

Email or Username

Password

☐ Remember me [Forgot password?](#)

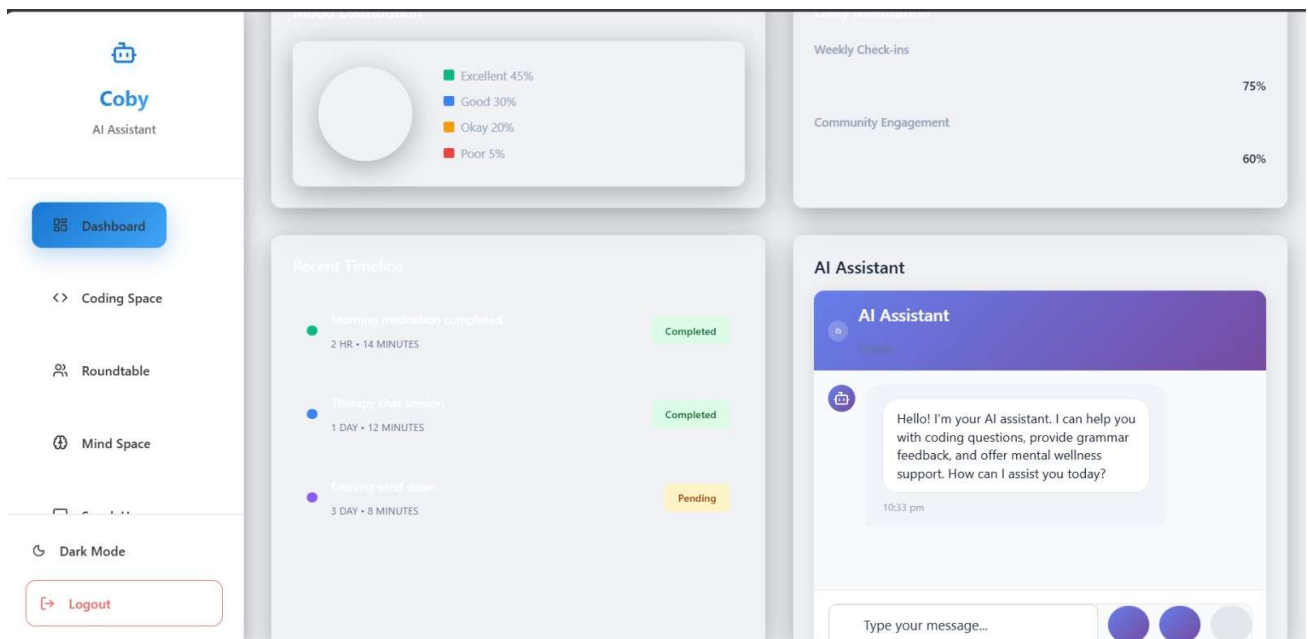
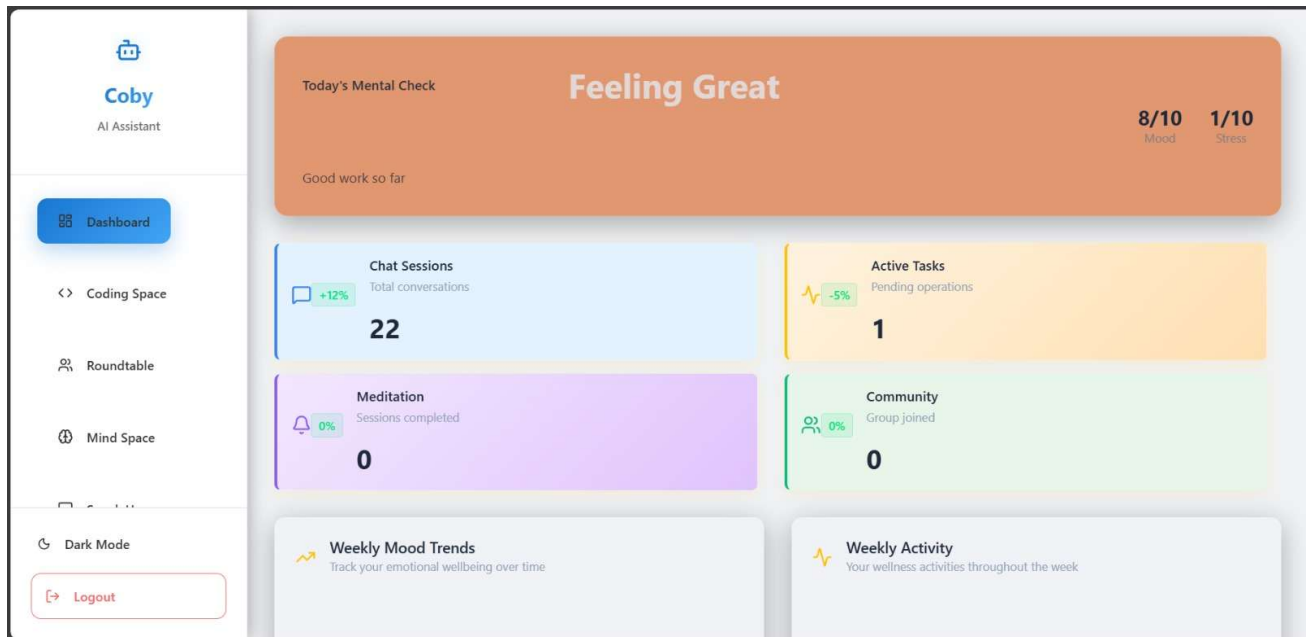
Sign In

or

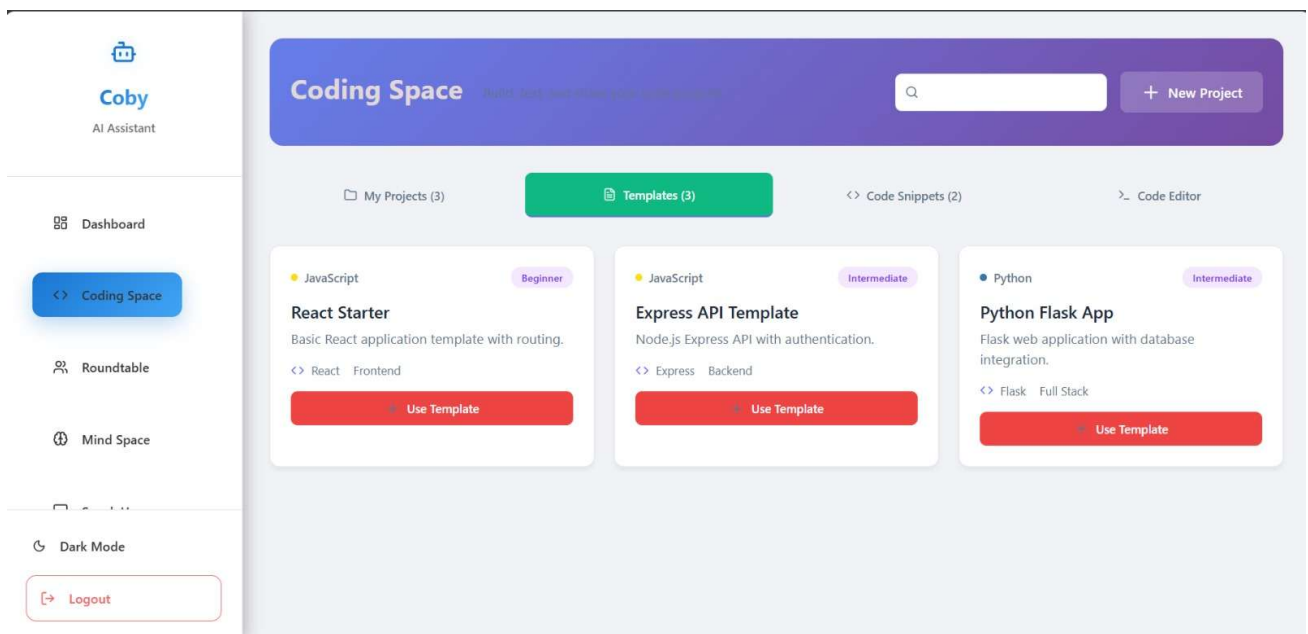
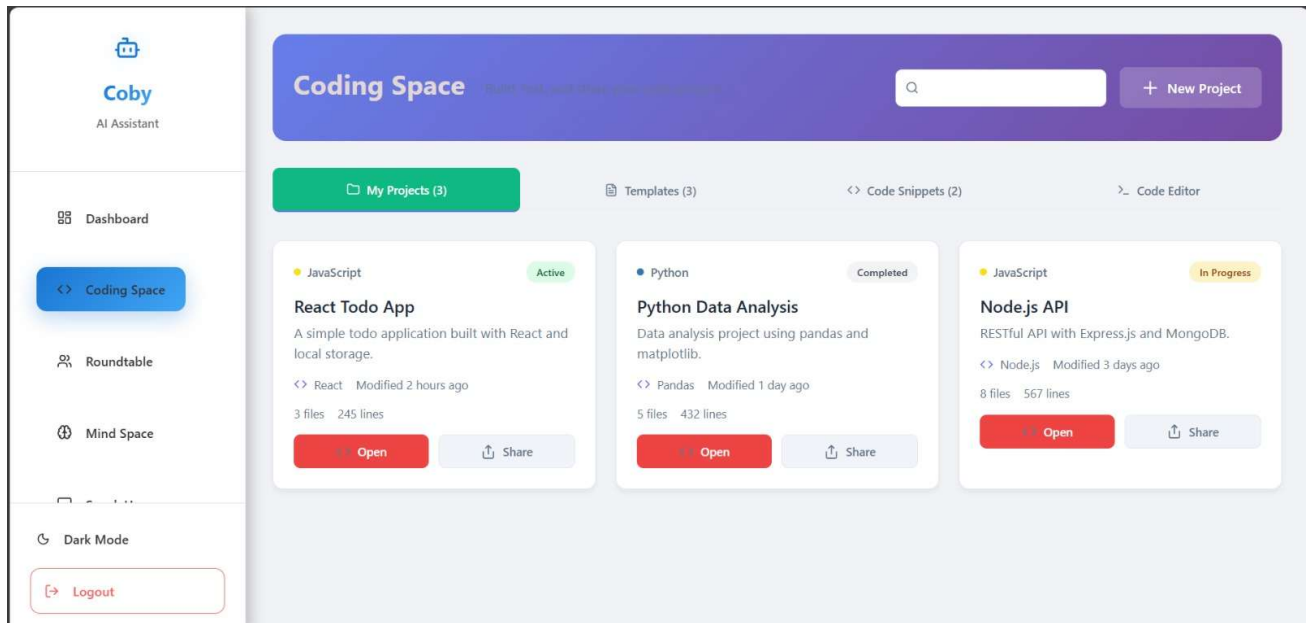
 Continue with Google

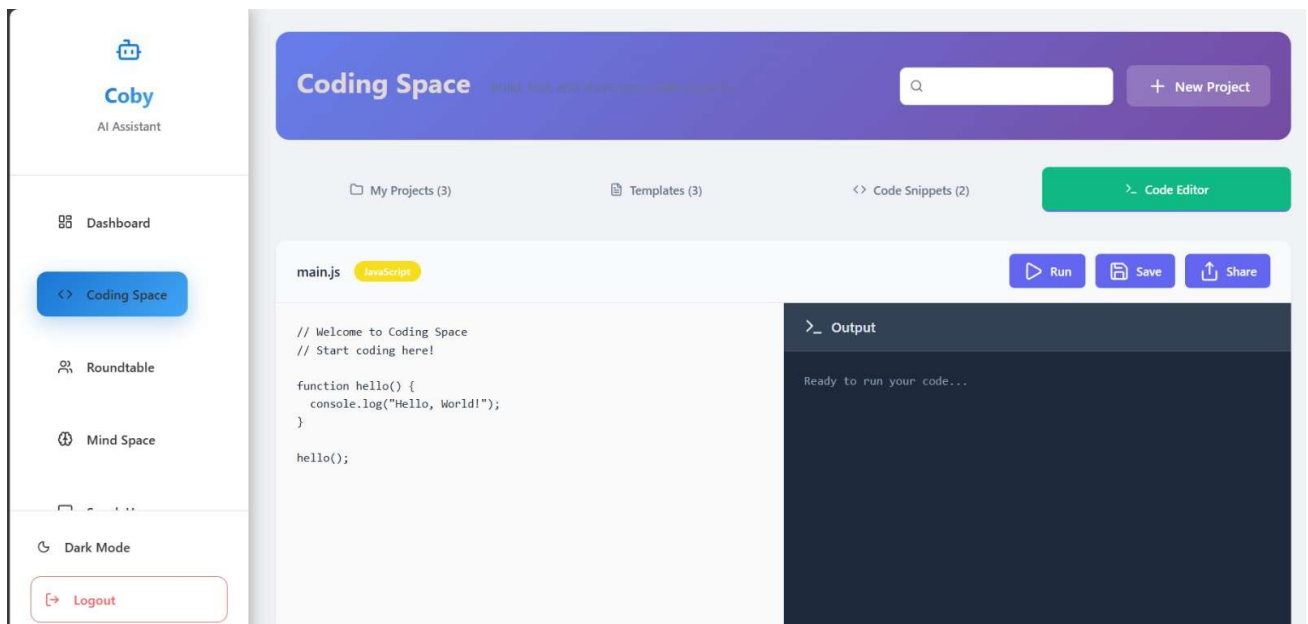
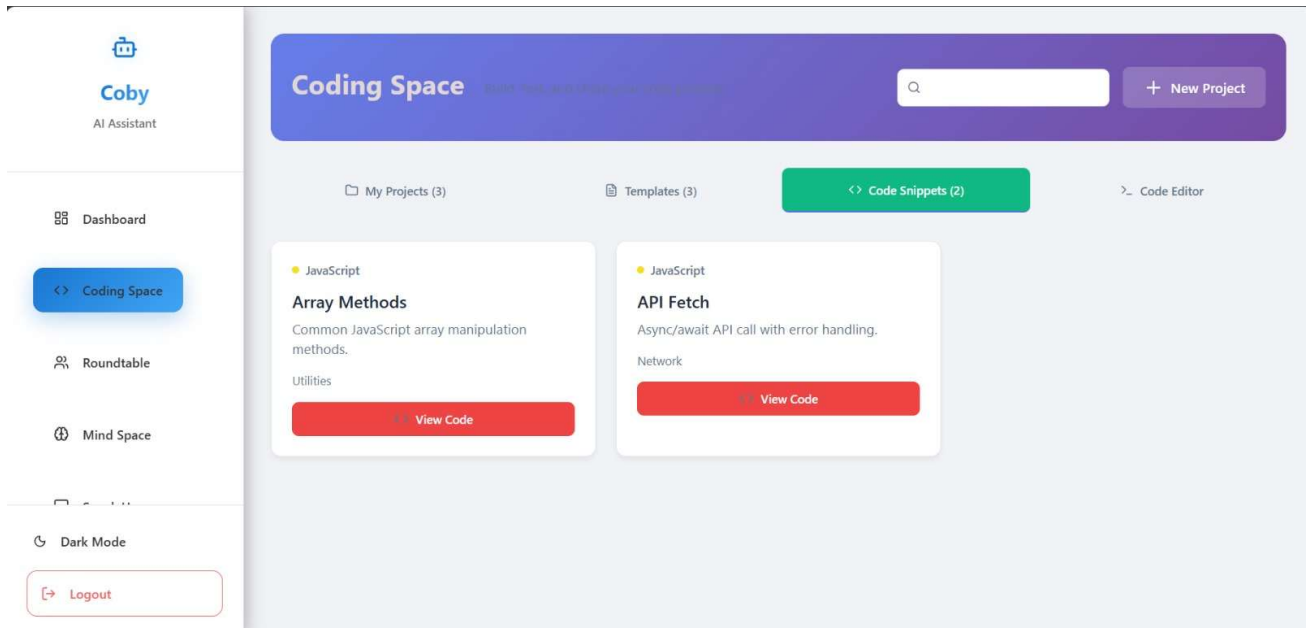
Don't have an account? [Sign up](#)

DASHBOARD:

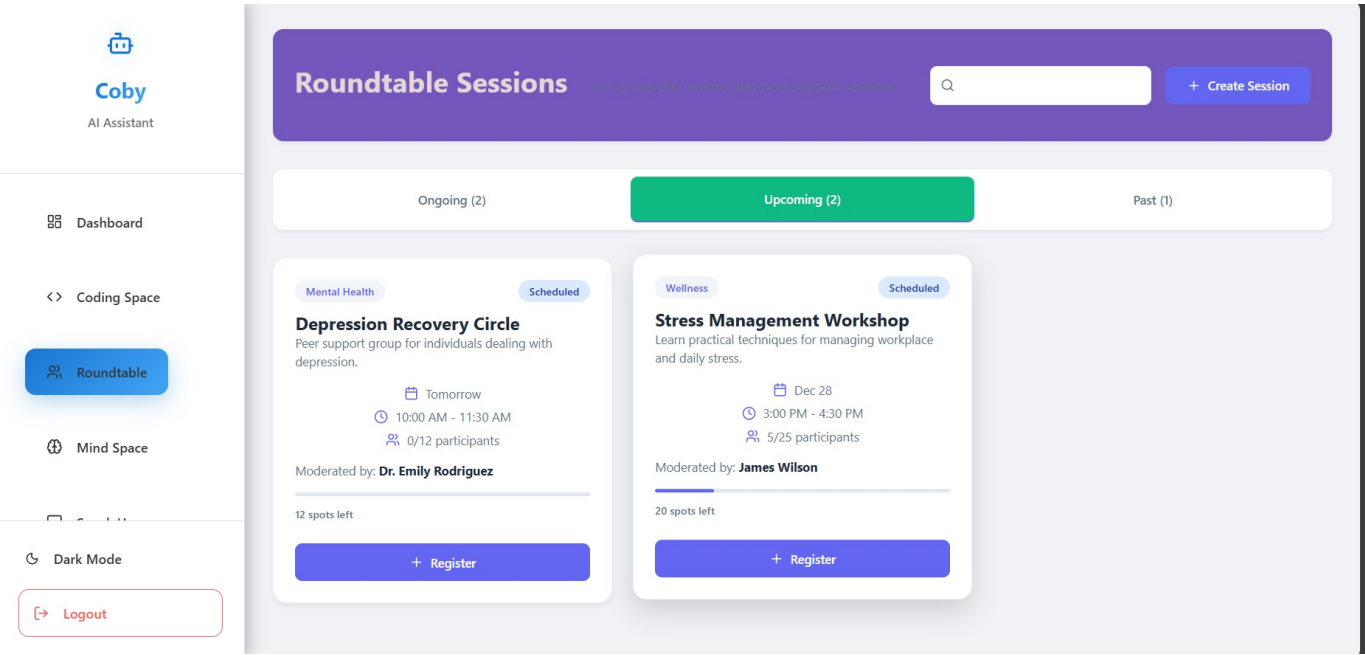
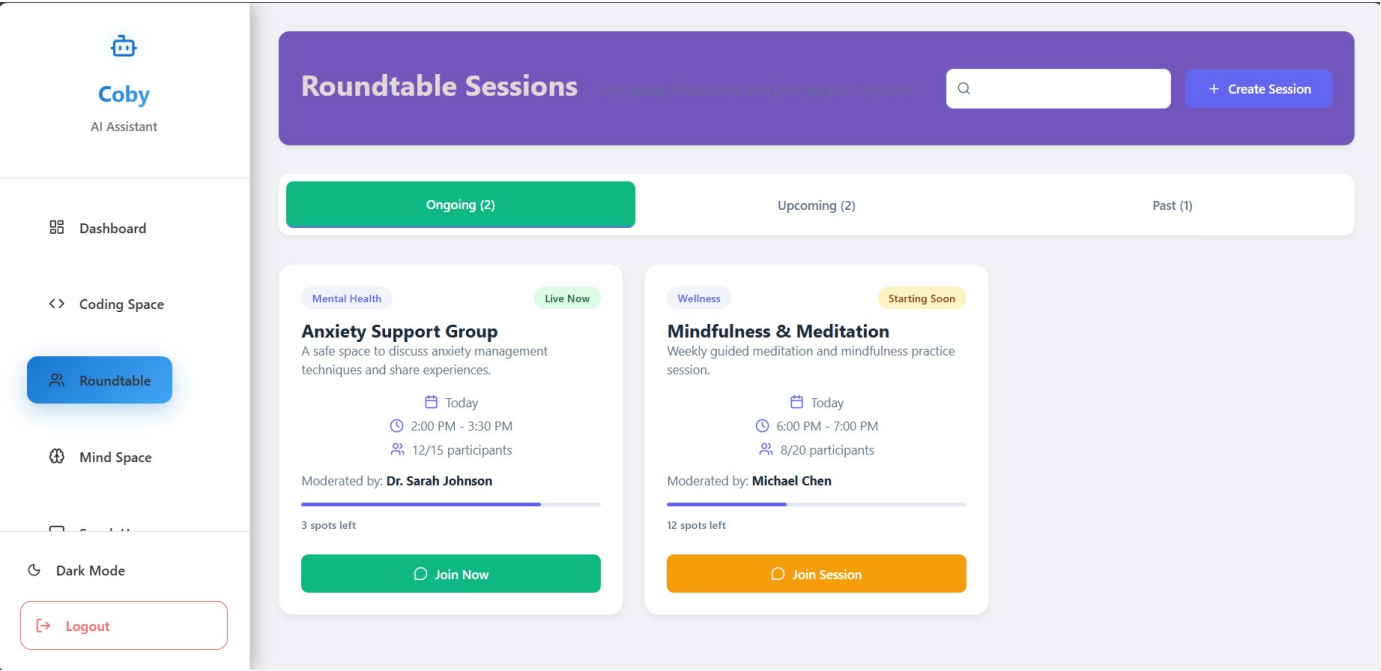


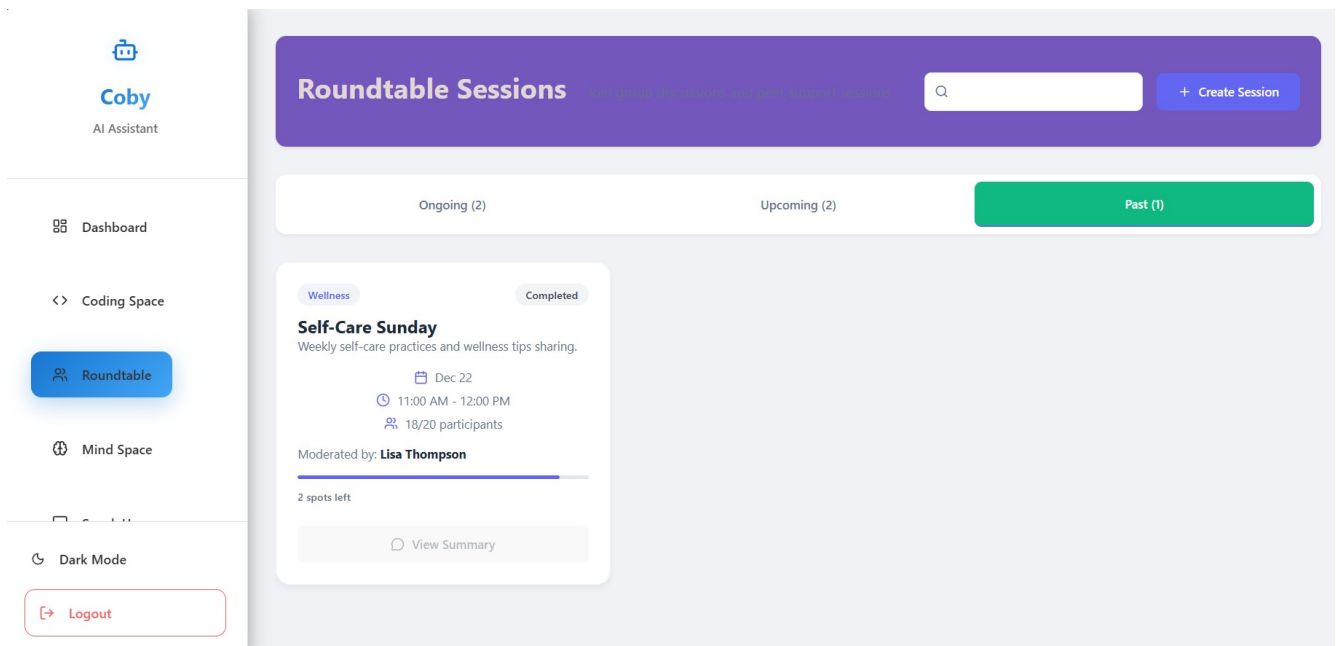
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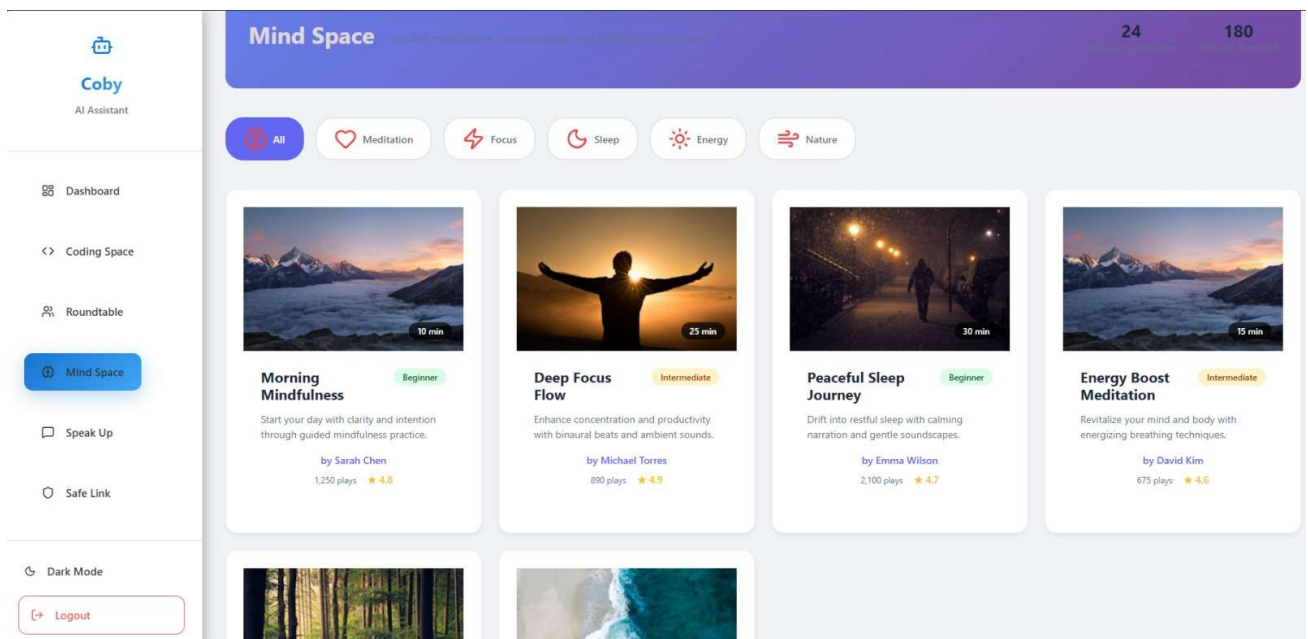


ROUNDTABLE SESSION:

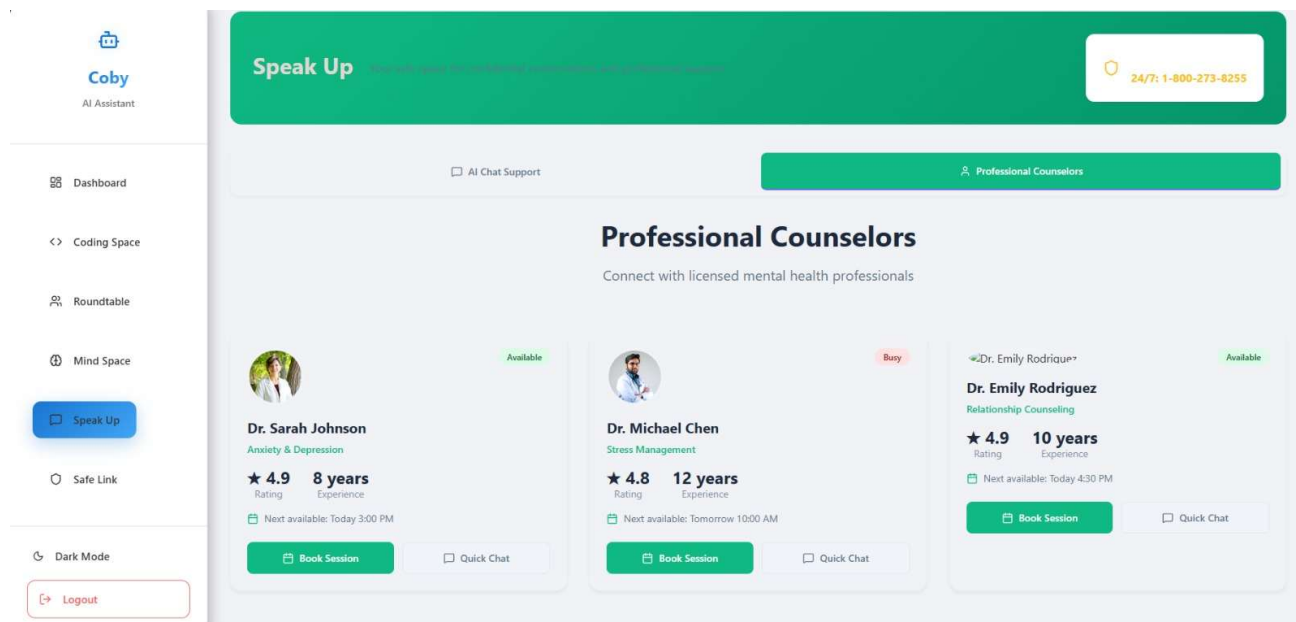
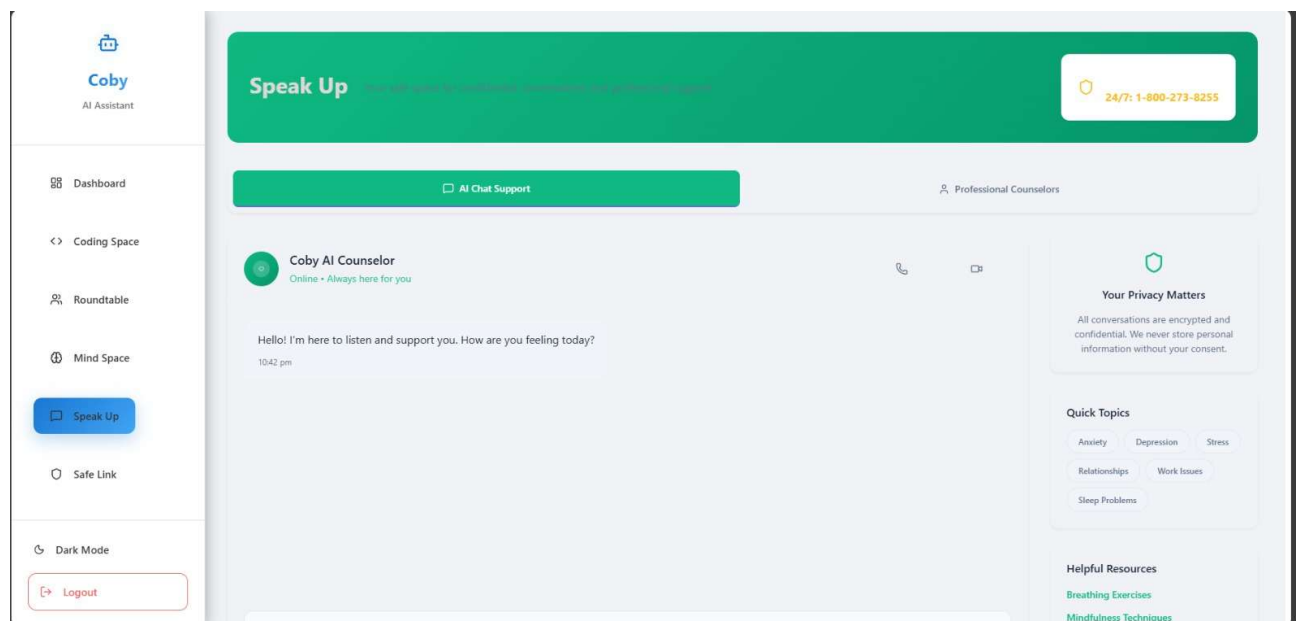




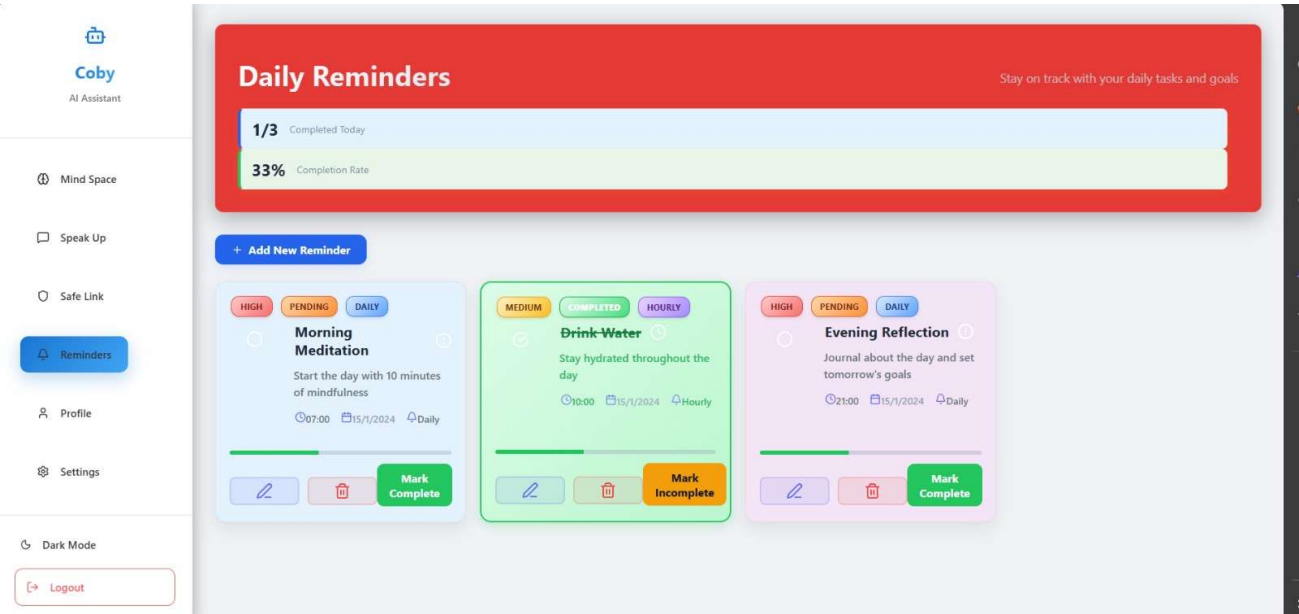
MINDSPACE:



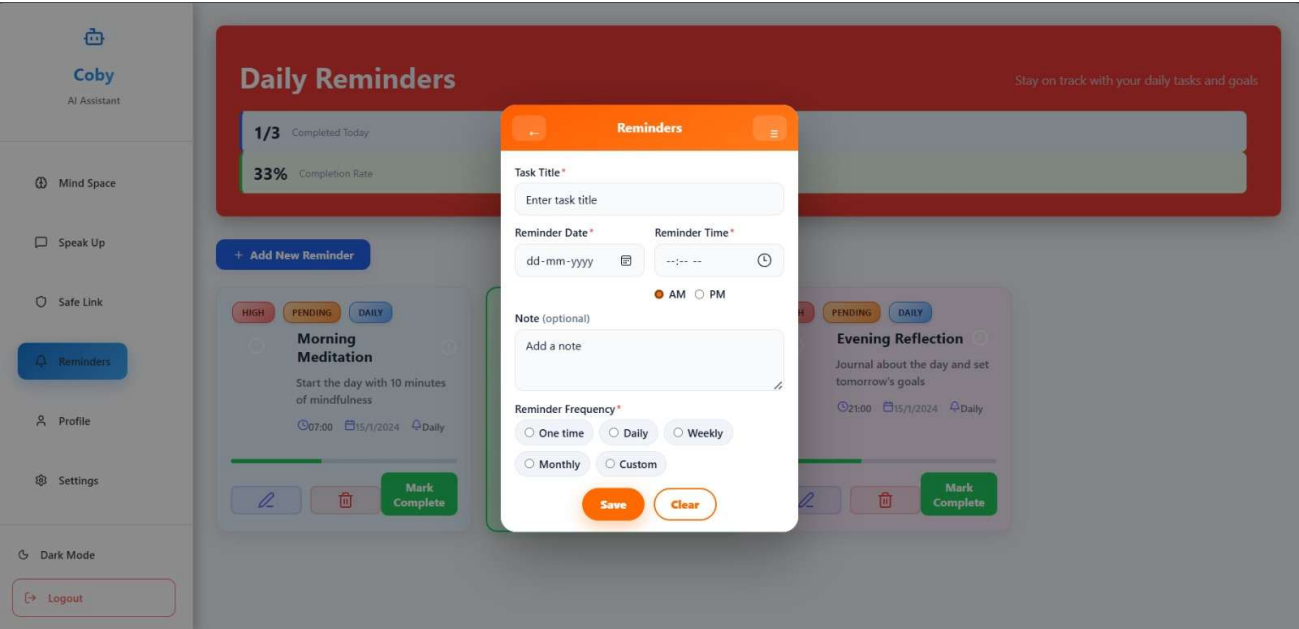
SPEAK UP:



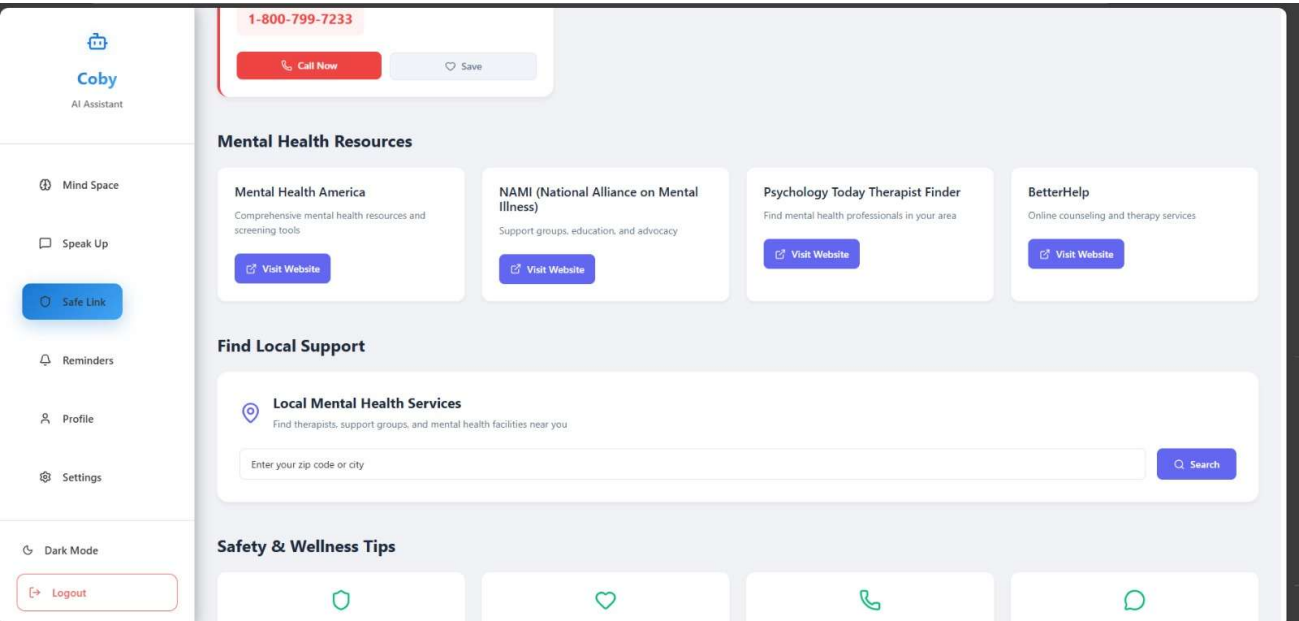
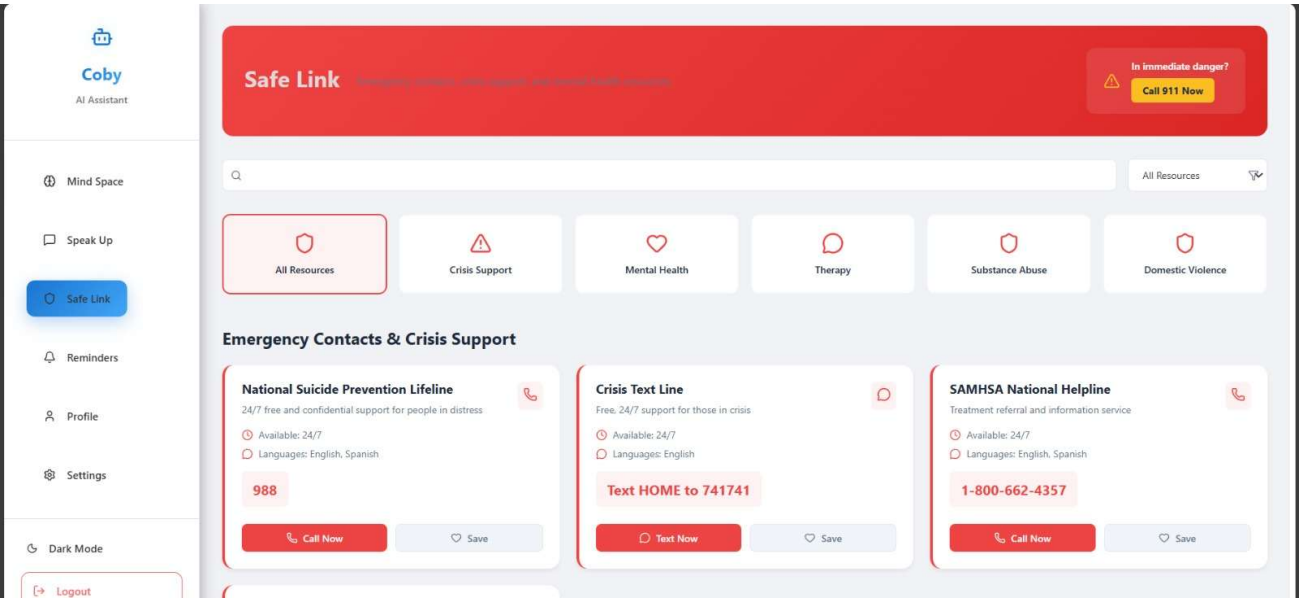
REMAINDER:



ADD REMAINDER:



SAFE LINK:



PROFILE:



Coby

AI Assistant

Mind Space

Speak Up

Safe Link

Reminders

Profile

Settings

Dark Mode

Logout



XXXX

Mental health advocate and wellness enthusiast.
Passionate about helping others on their journey to better mental health.

New York, NY

Member since Jan 2024

Edit Profile

Your Progress

45

Days Active

12

Goals Completed

7 days

Mood Streak

28

Sessions

Personal Information

Full Name

XXXX

Email Address

XXXX@example.com

Phone Number

+1 (555) 123-4567

Date of Birth

15/5/1990

Location

Achievements

First Steps

Completed your first wellness session

Earned 2024-01-15

Consistent Tracker

Logged mood for 7 consecutive days

Earned 2024-01-22

Community Helper

Participated in 5 group sessions

3/20

Mindfulness Master

Completed 20 meditation sessions

15/20

Recent Activity

7 days

Mood Streak

28

Sessions

Personal Information

Full Name

XXXX

Email Address

XXXX@example.com

Phone Number

+1 (555) 123-4567

Date of Birth

15/5/1990

Location

New York, NY

Bio

Mental health advocate and wellness enthusiast. Passionate about helping others on their journey to better mental health.

Emergency Contact

Contact Name

Jane Doe

Relationship

Sister

Phone Number

+1 (555) 987-6543

Consistent Tracker

Logged mood for 7 consecutive days

Earned 2024-01-22

Community Helper

Participated in 5 group sessions

3/20

Mindfulness Master

Completed 20 meditation sessions

15/20

Recent Activity

Morning Mindfulness

2024-01-25 10 min

Mood Check-in

2024-01-25 Good (8/10)

Anxiety Support Group

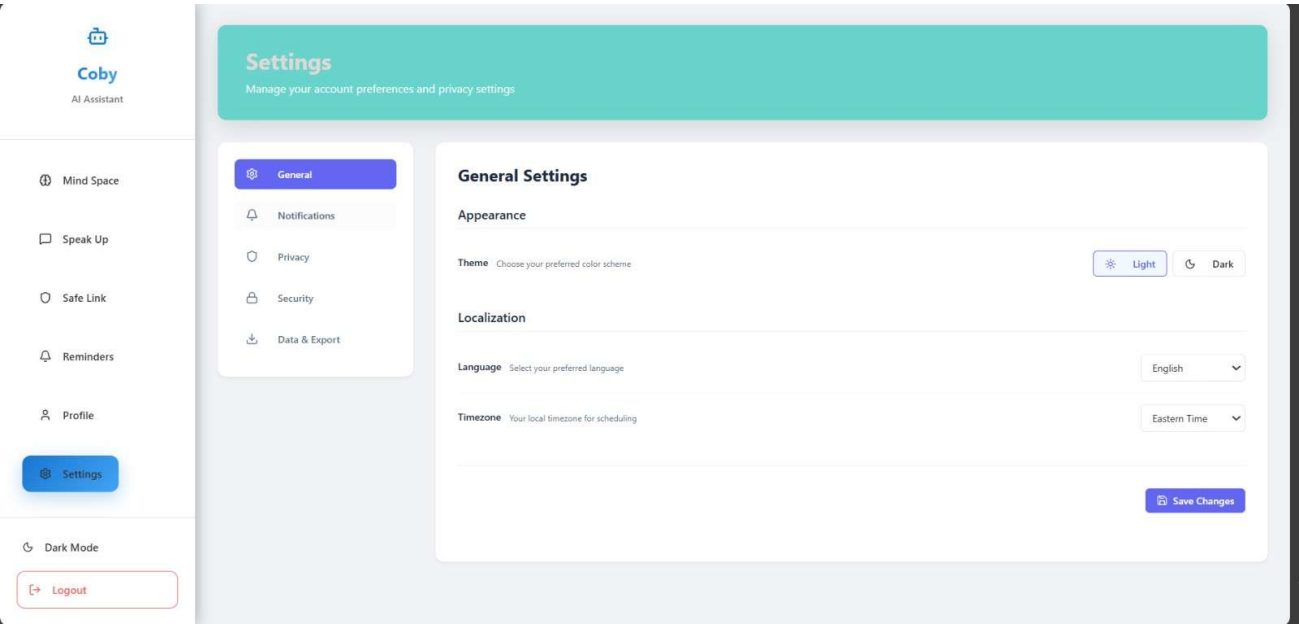
2024-01-24 60 min

Daily Exercise Goal

2024-01-24 completed

32

SETTINGS:



Conclusion

The AI Growth Chatbot Project shows how web development, automation, and artificial intelligence can be combined to create a powerful platform that makes everyday tasks easier for professionals and students. This system combines features like secure authentication, document summarisation, reminders, notifications, and AI-powered conversations to create a flexible productivity assistant. We obtained practical experience in full-stack development, API integration, and deployment readiness through this project. We created a scalable, secure, and flexible solution by utilising Flask for the backend, React.js for the frontend, MongoDB Atlas for cloud-based storage, and AI models from OpenAI Groq. By providing speech-to-speech for accessibility, this chatbot addresses the problem of utilising multiple apps for learning, reminders, and enquiries. This project shows how cutting-edge AI technologies can be integrated with reliable backend systems to deliver practical solutions.

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14. SpeechRecognition Python Library – <https://pypi.org/project/SpeechRecognition/>
15. PyPDF2 Library – <https://pypi.org/project/PyPDF2/>
16. Authlib Documentation – <https://docs.authlib.org/>
17. Requests Python Library – <https://docs.python-requests.org/>
18. LanguageTool API – <https://languagetool.org/dev>
19. Cloud Deployment Guidelines (Vercel, Netlify, Render) – Official Documentation
20. Various Stack Overflow discussions and GitHub repositories for troubleshooting and optimization.

PO & PSO Attainment

PO.No	Graduate Attribute	Attained	Justification
PO 1	Engineering knowledge	Yes	Applied core concepts of AI, NLP, and software engineering to design and implement a chatbot system
PO 2	Problem analysis	Yes	Analyzed student queries, identified functional requirements, and structured solutions to address communication gaps.
PO 3	Design/Development of solutions	Yes	Designed an AI-powered chatbot with real-time response capabilities to solve academic and administrative doubts.
PO 4	Conduct investigations of complex problems	Yes	Researched NLP models, AI frameworks, and dataset integration to create a functional, scalable solution.
PO 5	Modern Tool usage	Yes	Used AI/ML libraries, APIs, and deployment tools to develop and host the chatbot.
PO 6	The Engineer and society	Yes	Supports students and faculty by improving communication and reducing manual workloads in educational institutions.
PO 7	Environment and Sustainability	No	Not directly related to environmental or sustainability challenges.
PO 8	Ethics	Yes	Followed responsible AI practices, ensuring data privacy and ethical response generation.
PO 9	Individual and team work	Yes	Collaborated with peers and mentors to build, test, and refine the chatbot, showcasing teamwork.
PO 10	Communication	Yes	Enhanced communication by developing a system that simplifies student–faculty interaction.

PO.No	Graduate Attribute	Attained	Justification
PO 11	Project management and finance	Yes	Planned project phases (design, implementation, testing) and optimized resource usage effectively.
PO 12	Life-long learning	Yes	Explored new AI concepts, chatbot frameworks, and cloud services, demonstrating continuous learning.

PSO.No	Graduate Attribute	Attained	Justification
PSO 1	To analyze, design and develop solutions by applying the concepts of Robotics for societal and industrial needs.	No	The project focuses on AI and NLP, not robotics applications.
PSO 2	To create innovative ideas and solutions for real time problems in Manufacturing sector by adapting the automation tools and technologies.	No	Not related to manufacturing automation but applies automation in the education domain.